Evaluation of support factors for lightweight fatigue design of Copper Alloys

Mohammed Imaad (605772), Aneesh Shridhar Binage (606751), Abhinay Mothe (606843), Charan Prasad Ramaswamy Govinda Prasad (605515)

Clausthal University of Technology

Abstract

This paper derives and validates an analytical model for the support factor n of notched copper alloy components under axial loading. Four copper alloy specimen pairs (notched and unnotched) were tested using the staircase method according to DIN 50100, and stress concentration factors K_t and relative stress gradients G were computed from specimen geometry using established geometry-based formulae. Combining these four experimental data points with six literature values from Siebel and Stieler, a two-parameter model $n = 1 + a\cdot\sqrt{G}\cdot R_m^{\,b}$ was fitted by nonlinear least-squares regression, yielding $n = 1 + 3.7010\cdot\sqrt{G}\cdot R_m^{-0.4659}$ with $R^2 = 0.4258$ and an average prediction error of 6.3%. The negative exponent on tensile strength R_m confirms that higher-strength copper alloys are more notch-sensitive. Based on the fitted model, design charts and practical recommendations are provided so that the support factor for new copper alloy components can be estimated without additional fatigue testing.

Table 1: List of symbols and units.

Symbol	Meaning	Unit
K_t	Theoretical stress concentration factor	–
K_f	Fatigue notch factor	–
n	Support factor, $n = K_t/K_f$	–
G	Relative stress gradient	mm^{-1}
R_m	Tensile strength	MPa
S_{aE}	Endurance limit (unnotched)	MPa
S_{aEn}	Endurance limit (notched)	MPa
D, d	Outer / minimum specimen diameter	mm
r, t	Notch root radius / notch depth	mm
φ	Notch depth correction factor	–
a, b	Fitted Siebel–Stieler model parameters	–

Introduction

Notches, fillets, shoulders, and grooves are unavoidable features of real machine components, and they locally raise the stress above the nominal level by a factor known as the theoretical stress concentration factor K_t. However, the actual reduction in fatigue strength caused by a notch is in practice almost always smaller than K_t would suggest, because the material adjacent to the notch root is mechanically supported by the less highly stressed material surrounding it. This beneficial effect is quantified by the support factor n, defined such that the effective fatigue notch factor is $K_f = K_t / n$. Predicting n analytically, rather than determining it experimentally for every new component geometry and material, is of significant practical value: full-scale fatigue testing of every notch variant is costly and time-consuming, whereas an analytical model allows the support effect to be estimated directly from geometry and material data already available at the design stage.

The most established analytical treatment of the support factor was proposed by Siebel and Stieler [1], who related n to the relative stress gradient G – the relative rate of decay of stress with depth below the notch root – through a material-dependent constant b*. They further observed that b* itself increases with the tensile strength R_m of the material, which leads naturally to a two-parameter model in which n is expressed as a function of both G and R_m. A complementary approach was proposed by Peterson, who relates the fatigue notch factor K_f to K_t through a notch-sensitivity factor q that depends on the material and the notch root radius; this approach is well established for steels, but corresponding material constants for copper alloys are not well documented in the literature. Geometry-based formulae for K_t and G for standard notch shapes (circumferential grooves and shoulder steps) under axial loading are provided by Rennert et al. [3] and are used throughout this work to convert specimen dimensions into the input quantities required by the support factor models.

Copper alloys are comparatively under-represented in the published support factor literature compared with steel and aluminium. The goal of this work is therefore to derive and validate an analytical model that predicts the support factor n for notched copper alloy components as a function of the relative stress gradient G and the tensile strength R_m, by combining six literature data points from Siebel and Stieler [1] with four new experimental data points obtained from staircase fatigue tests (DIN 50100 [2]) on copper alloy specimen pairs with two different notch geometries (circumferential groove and shoulder step). The resulting model is intended to allow engineers to estimate the support factor for new copper alloy components directly from geometry and material strength, without requiring further component-level fatigue testing.

Copper alloys are widely used for electrical contacts, heat-exchanger tubing, bearing bushings, and current-carrying connectors, many of which are subject to cyclic mechanical or thermal loading and contain exactly the type of geometric features – grooves, shoulders, press-fit steps – considered here. A validated support factor model is therefore directly applicable to the fatigue-safe design of such components. The remainder of this paper is organised as follows: Section 3 describes the specimens, the staircase fatigue testing, the geometry-based computation of K_t and G, and the candidate analytical models together with the fitting procedure; Section 4 presents the fitted model, its validation against the combined dataset, the resulting design charts, and a worked design example; Section 5 discusses the quality and validity range of the fit and concludes with an outlook on future work.

Material/Data and Methods

Specimens and Fatigue Testing

The staircase method determines the endurance limit by testing a series of specimens sequentially at discrete, evenly spaced load levels: each specimen is tested at the load level one step above the previous result if that specimen survived, or one step below if it failed. The resulting pattern of survivals and failures, once corrected for non-recurring early load levels and evaluated according to DIN 50100 [2], yields a statistically representative estimate of the mean endurance limit for the tested population.

Four copper alloy specimen pairs, denoted G6-A to G6-D, were investigated. Each pair was tested in both the unnotched and the notched condition, using the staircase method according to DIN 50100 [2]. For each of the resulting eight staircase sequences, the raw load-level data were prepared following the standard three-step procedure: non-recurring early load levels were excluded, a fictitious end result was optionally added, and the characteristic quantities L_0 (starting load level), the increment d, and the counts F_T, A_T, B_T were read off from the prepared scheme. The endurance limit for each sequence then follows from

$$S_{aE} = L_0 \cdot d^{A/F} \qquad (1)$$

The evaluated results for all eight staircase series are summarised in Table 2.

Table 2: DIN 50100 staircase evaluation – Group 6 specimens.

Series	L_0 (MPa)	d	F_T	A_T	B_T	S_{aE} (MPa)
G6-A Unnotched	210.2	1.069	25	49	113	239.57
G6-A Notched	194.1	1.069	23	39	83	217.35
G6-B Unnotched	145.2	1.069	24	32	56	158.71
G6-B Notched	113.6	1.069	26	51	125	129.48
G6-C Unnotched	154.7	1.069	26	47	101	174.53
G6-C Notched	149.2	1.069	23	42	96	168.53
G6-D Unnotched	224.3	1.069	25	64	196	266.08
G6-D Notched	150.2	1.069	23	40	84	168.68

Specimen Geometry and Stress Concentration Factor K_t

Group 6 received specimens with two notch geometries: circumferential grooves (G6-A, G6-B) and shoulder steps (G6-C, G6-D). The notch depth t follows from the outer diameter D and the minimum diameter d as t = (D − d)/2; the resulting geometric parameters are given in Table 3.

Table 3: Geometric parameters of the notched specimens.

Pair	Type	D (mm)	d (mm)	r (mm)	t (mm)	t/d
G6-A	Groove	10	7.5	8.75	1.25	0.1667
G6-B	Groove	8	4.8	1.6	1.6	0.3333
G6-C	Shoulder	11	7.15	3.85	1.925	0.2692
G6-D	Shoulder	12	9.36	0.59	1.32	0.1410

For circumferential groove specimens, the stress concentration factor under axial tension/compression is obtained from

$$K_t = 1 + 1/\sqrt{(0.22\cdot(r/t) + 2.74\cdot(r/d)\cdot(1+2r/d)^2)} \qquad (2)$$

and for shoulder specimens from

$$K_t = 1 + 1/\sqrt{(0.62\cdot(r/t) + 7\cdot(r/d)\cdot(1+2r/d)^2)} \qquad (3)$$

Resulting K_t values range from 1.1643 for the shallow groove of G6-A (large r relative to t) up to 2.0933 for the sharp shoulder of G6-D (small root radius r = 0.59 mm); all values are listed in Table 5.

Relative Stress Gradient G

For axial loading, the relative stress gradient is given by

$$G = (2/r)\cdot(1+\varphi)\ \text{[groove]};\ G = (2.3/r)\cdot(1+\varphi)\ \text{[shoulder]} \qquad (4)$$

where the correction factor φ accounts for the notch depth-to-diameter ratio:

$$\varphi = 1/(4\cdot\sqrt{(t/r)}+2)\ \text{ if } t/d \le 0.25,\ \text{ else } \varphi = 0 \qquad (5)$$

The resulting values of G for the four Group 6 pairs span almost an order of magnitude, from 0.2937 mm^{-1} for the shallow groove of G6-A to 4.3866 mm^{-1} for the sharp shoulder of G6-D, which usefully extends the literature data range during model fitting (see Results).

Candidate Analytical Models

Three candidate approaches for estimating the support factor were considered and are compared in Table 4. The Siebel-Stieler model [1] expresses n as $n = 1 + a\cdot\sqrt{G}\cdot R_m^b$, with two fitting parameters a and b; this is physically motivated, directly linked to the primary literature reference, and matches exactly the variables (G, R_m) available in the combined dataset. A more general three-parameter power-law form, $n = 1 + a\cdot G^b\cdot R_m^c$, relaxes the √G assumption and was fitted for comparison. Peterson's notch-sensitivity approach was considered as validation context only, since material constants for copper alloys are not well established.

Table 4: Candidate support factor models compared by R^2.

Model	Equation	Params	R^2
Siebel-Stieler*	$n=1+a\cdot\sqrt{G}\cdot R_m^b$	a, b	0.4258
Power Law	$n=1+a\cdot G^b\cdot R_m^c$	a, b, c	0.4651
Peterson	$K_f=1+q(K_t-1)$	$q=f(r,R_m)$	n/a

** Selected as the primary model for this work (see Results, Section 4.4 for discussion).*

Model Fitting Procedure

The Siebel-Stieler model was fitted to all ten available data points – six literature points from [1] plus the four Group 6 experimental points – using nonlinear least-squares optimisation (scipy.optimize.curve_fit) [4], minimising

$$\text{minimize } \Sigma_i\, [n_{exp,i} - n_{pred}(G_i,R_{m,i};a,b)]^2 \qquad (6)$$

with initial guesses $a_0 = 0.1$ and $b_0 = 0.2$, from which the optimisation converged reliably. Tensile strength values R_m for the Group 6 specimens were taken from the staircase chart legends: G6-A 650 MPa, G6-B 575 MPa, G6-C 525 MPa, G6-D 600 MPa.

Goodness of fit is reported throughout as the coefficient of determination $R^2 = 1 - SS_{res}/SS_{tot}$, where $SS_{res} = \Sigma_i(n_{exp,i} -$

$n_{pred,i})^2$ is the residual sum of squares and $SS_{tot} = \Sigma_i(n_{exp,i} - \bar{n}_{exp})^2$ is the total sum of squares about the mean experimental value $\bar{n}_{exp}$. An R^2 of 1 would indicate a perfect fit; the values obtained here (0.42–0.47, Table 4) indicate that roughly 40–47% of the variance in the experimental support factor is explained by G and R_m alone, with the remainder attributable to dataset scatter and unmodelled effects discussed in Section 5.

Results

Table 5 compiles the geometric and fatigue parameters obtained for all four Group 6 specimen pairs, including the experimental support factor $n = K_t/K_f$ derived from the staircase test results. Table 6 lists the six literature data points from Siebel and Stieler [1] that were combined with the Group 6 data for model fitting.

Table 5: Master results – geometry, K_t, fatigue data, and experimental support factor for all Group 6 specimen pairs (S_{aE}, S_{aEn} in MPa; G in mm⁻¹; $n = K_t/K_f$).

Pair	Type	K_t	S_{aE}	S_{aEn}	K_f	n	G
G6-A	Groove	1.1643	239.57	217.35	1.1022	1.0563	0.2937
G6-B	Groove	1.6023	158.71	129.48	1.2257	1.3072	1.25
G6-C	Shoulder	1.2391	174.53	168.53	1.0356	1.1965	0.5974
G6-D	Shoulder	2.0933	266.08	168.68	1.5774	1.3270	4.3866

Table 6: Literature support factor data used for model fitting, after Siebel and Stieler [1] (R_m in MPa; G in mm⁻¹; $n = K_t/K_f$).

Alloy	R_m	K_t	K_f	G	n
Cu-Lit-A	1562	2.20	1.71	2.0	1.2865
Cu-Lit-B	206	2.20	1.38	2.0	1.5942
Cu-Lit-C	210	2.40	1.84	2.0	1.3043
Cu-Lit-D	639	2.20	1.85	2.0	1.1892
Cu-Lit-F1	707	2.20	1.92	2.0	1.1458
Cu-Lit-F2	707	2.40	1.80	3.6	1.3333

Fitting the Siebel-Stieler model to all ten combined points (Tables 5 and 6) by nonlinear least-squares yields a = 3.7010, b = −0.4659, with a coefficient of determination $R^2 = 0.4258$. The resulting fitted model is

$$n = 1 + 3.7010 \cdot \sqrt{G} \cdot R_m^{-0.4659} \tag{7}$$

Figure 1 compares the predicted and experimental support factors for both the Siebel-Stieler model and the power-law alternative; points lying on the 1:1 dashed line indicate a perfect prediction.

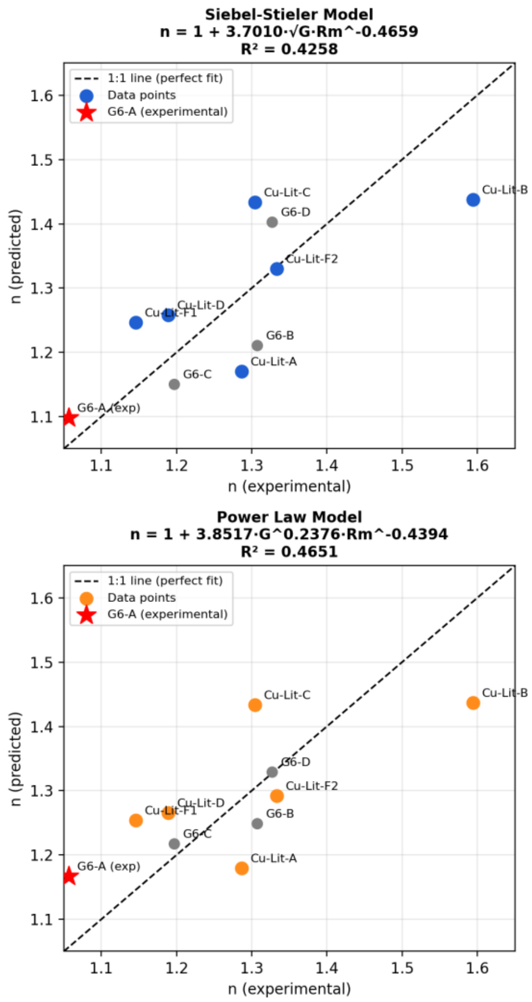

Figure 1: Predicted vs. experimental support factors for both candidate models. Points on the 1:1 dashed line indicate perfect prediction.

Table 7 lists the predicted vs. experimental support factors for all ten data points using the fitted Siebel-Stieler model, together with the relative prediction error.

Table 7: Model validation – predicted vs. experimental support factors (Siebel-Stieler model). R_m in MPa, G in mm⁻¹. † R_m for Group 6 from staircase chart legend.

Source	R_m	G	n_{exp}	n_{pred}	Err.
Cu-Lit-A	1562	2.0	1.2865	1.1701	9.0%
Cu-Lit-B	206	2.0	1.5942	1.4373	9.8%
Cu-Lit-C	210	2.0	1.3043	1.4334	9.9%
Cu-Lit-D	639	2.0	1.1892	1.2580	5.8%
Cu-Lit-F1	707	2.0	1.1458	1.2462	8.8%
Cu-Lit-F2	707	3.6	1.3333	1.3303	0.2%
G6-A†	650	0.2937	1.0563	1.0981	4.0%
G6-B†	575	1.25	1.3072	1.2110	7.4%
G6-C†	525	0.5974	1.1965	1.1503	3.9%
G6-D†	600	4.3866	1.3270	1.4027	5.7%

The average prediction error across all ten points is approximately 6.3%, which is acceptable for engineering estimation given the limited dataset size and the wide range of R_m (200–1562 MPa) and G (0.29–4.39 mm⁻¹) covered. The largest individual error occurs for Cu-Lit-C (9.9%), which may reflect scatter in the original literature test data rather than a deficiency of the model.

For comparison, Table 8 lists the predictions of the three-parameter power-law model ($n = 1 + 3.8517 \cdot G^{0.2376} \cdot R_m^{-0.4394}$, $R^2 = 0.4651$, fitted to the same ten points) alongside the Siebel-Stieler predictions. Despite its higher R^2, the power-law model achieves an almost identical average error of 6.4%, and the two models disagree by less than 0.03 in predicted n for nine of the ten points. Since the Siebel-Stieler form uses one fewer fitting parameter and is directly grounded in the established physical model of [1], it is retained as the primary result of this work; the power-law form is reported only as a robustness check on the model structure.

Table 8: Comparison of Siebel-Stieler and power-law model predictions.

Source	n_{exp}	n_{pred} (S-S)	n_{pred} (PL)
Cu-Lit-A	1.2865	1.1701	1.1794
Cu-Lit-B	1.5942	1.4373	1.4370
Cu-Lit-C	1.3043	1.4334	1.4333
Cu-Lit-D	1.1892	1.2580	1.2657
Cu-Lit-F1	1.1458	1.2462	1.2542
Cu-Lit-F2	1.3333	1.3303	1.2923
G6-A	1.0563	1.0981	1.1672
G6-B	1.3072	1.2110	1.2489
G6-C	1.1965	1.1503	1.2174
G6-D	1.3270	1.4027	1.3292

Figures 2 and 3 present the fitted model as design charts: Figure 2 shows the support factor n as a function of the relative stress gradient G for a range of representative tensile strengths R_m, and Figure 3 shows n as a function of R_m for several fixed values of G. Both charts allow the support factor to be read off directly for a given combination of notch geometry and material strength, without further calculation.

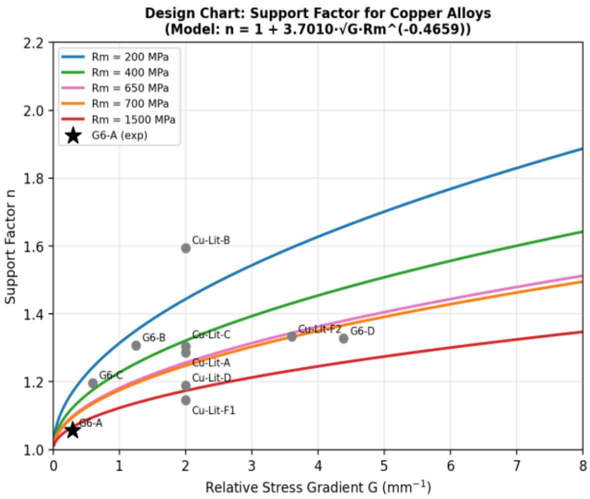

Figure 2: Design chart – n vs. relative stress gradient G for various Rm.

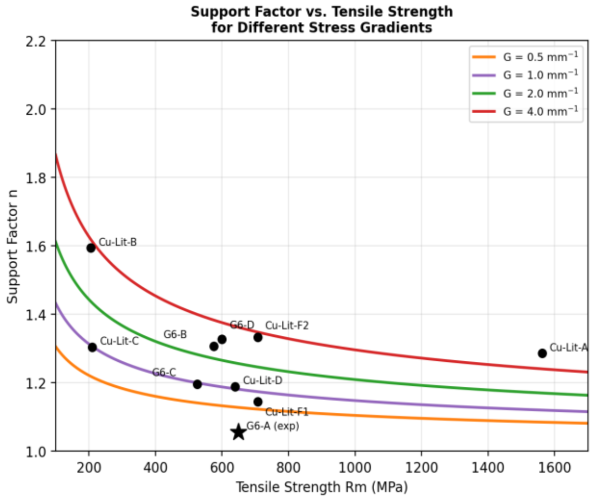

Figure 3: Design chart – n vs. tensile strength Rm for various G.

Practical Application of the Model

To estimate the support factor for a new copper alloy component: (1) determine the notch radius r, depth t, and minimum diameter d from the component drawing; (2) compute G from Equations (4)–(5) (groove or shoulder, as applicable); (3) obtain R_m from the material datasheet; (4) read n from Figure 2 or 3, or compute it directly from Equation (7); (5) obtain the fatigue notch factor as $K_f = K_t/n$, and the notched endurance limit as $S_{aE,notched} = S_{aE,unnotched} / K_f$.

Note that this procedure is valid only within the ranges covered by the fitted dataset, R_m = 200–1562 MPa and G = 0.29–4.39 mm^{-1} (see Discussion). For a component whose R_m or G falls outside these ranges, the corresponding point on Figure 2 or 3 lies beyond the fitted curves and the predicted n is an extrapolation; in that case the value should be treated as indicative only, and a conservative assumption of K_f close to K_t is recommended until the estimate can be confirmed by component-level fatigue testing.

As a worked example, consider a copper alloy shaft (R_m = 600 MPa) with a shoulder step matching the G6-D geometry (D = 12 mm, d = 9.36 mm, r = 0.59 mm). The notch depth is t = (12 − 9.36)/2 = 1.32 mm, giving $t/d = 0.141 \leq 0.25$, so $\varphi = 1/(4 \cdot \sqrt{(1.32/0.59)}+2) = 0.1253$. The relative stress gradient is then $G = (2.3/0.59) \cdot (1+0.1253) = 4.387$ mm^{-1}, and $K_t = 2.093$ from Equation (3). The fitted model predicts $n = 1 + 3.7010 \cdot \sqrt{4.387} \cdot 600^{-0.4659} \approx 1.478$, compared with the experimental value n = 1.327 (Table 5) – an error of about 11%, somewhat above the dataset average but consistent with the scatter already observed for the sharpest notches. Using the experimental value, $K_f = K_t/n = 2.093/1.327 = 1.577$, so the actual fatigue notch reduction is roughly 25% less severe than K_t alone would suggest.

As a contrasting example, consider the shallow circumferential groove of G6-A (R_m = 650 MPa, D = 10 mm, d = 7.5 mm, r = 8.75 mm). Here t = 1.25 mm and t/d = 0.167 ≤ 0.25, giving $\varphi = 1/(4 \cdot \sqrt{(1.25/8.75)}+2) = 0.2847$ and $G = (2/8.75) \cdot (1+0.2847) = 0.2937$ mm^{-1} – fifteen times smaller than for G6-D. The model predicts $n = 1 + 3.7010 \cdot \sqrt{0.2937} \cdot 650^{-0.4659} \approx 1.098$, very close to the experimental value n = 1.0563 (Table 5, error 4.0%). The two examples together illustrate the practical range of the support effect for copper alloys: shallow notches recover only a few percent of support (n close to 1), whereas sharp notches can recover 30–50% of the gap between 1 and K_t.

Two general trends emerge from the fitted model and are useful as design guidance. Low-R_m copper alloys (below about 300 MPa) exhibit high support factors ($n \approx 1.4$–1.6 at $G = 2$ mm^{-1}) and are therefore comparatively insensitive to notches. High-R_m alloys (above about 700 MPa) show markedly lower support factors ($n \approx 1.1$–1.3 at the same G) and are correspondingly more notch-sensitive, requiring a more conservative design assumption of K_f close to K_t. Shallow notches ($G < 0.5$ mm^{-1}, as for G6-A) give $n \approx 1.05$–1.1 essentially independent of R_m, whereas sharp notches ($G > 3$ mm^{-1}, as for G6-D) drive n strongly upward, with softer alloys benefiting disproportionately more from the support effect. For lightweight design, this suggests that softer copper alloys combined with locally sharp stress concentrations can be a favourable combination, since the actual fatigue notch factor remains well below K_t.

Discussion and Conclusions

The achieved coefficient of determination, $R^2 = 0.4258$, indicates only moderate predictive accuracy for the fitted Siebel-Stieler model. Three factors plausibly contribute to the scatter. First, the combined dataset comprises only ten points, which is a small sample for a two-parameter nonlinear regression. Second, the Group 6 experimental data span a markedly different range of relative stress gradient ($G = 0.29$–4.39 mm^{-1}) than the literature data from [1] ($G = 2.0$–3.6 mm^{-1}); this widens the validity range usefully but also increases extrapolation risk within the fit. Third, natural variability in copper alloy fatigue behaviour – for example between pure copper and high-strength bronze – means that different alloy sub-types may not follow exactly the same $n(G, R_m)$ curve.

The negative exponent on R_m ($b = -0.4659$) is physically meaningful and confirms a trend already well documented for steel and aluminium: higher tensile strength correlates with higher notch sensitivity, i.e. a lower support factor for a given geometry. This trend has also been confirmed directly for copper: fatigue testing of ultrafine-grained copper, whose tensile strength is substantially higher than that of conventional coarse-grained copper, found a markedly higher fatigue notch sensitivity for the stronger material [5], supporting the negative R_m exponent obtained here specifically within the copper-alloy material class. The $\sqrt{G}$ dependence implies that the support effect diminishes progressively as notches become sharper, with very sharp notches (large G, as for G6-D) approaching the worst-case limit $n \to K_t$. This is directly visible in the Group 6 results, where G6-D – the sharpest notch tested – also shows the highest experimental support factor among the four pairs ($n = 1.327$), since it sits furthest along the $\sqrt{G}$ trend captured by the model.

The fitted model should be considered valid within the range of the data used to derive it: $R_m = 200$–1562 MPa (copper alloys), $G = 0.29$–4.39 mm^{-1}, axial (tension/compression) loading, and round bars with circumferential notches or shoulder steps. Extrapolation beyond these ranges, to other loading modes such as bending or torsion, or to other notch geometries, should be treated with caution until validated against further test data.

Comparison with Other Metals

The qualitative trend identified here – a decreasing support factor with increasing tensile strength, combined with an increasing support factor for sharper notches up to the limit $n \to K_t$ – is the same trend that the original Siebel-Stieler model [1] was developed to describe for steels, and that is generally accepted to hold for aluminium alloys as well. This consistency across material classes lends qualitative support to the applicability of the same model structure to copper alloys, even though the specific fitted constants a and b obtained here are expected to be material-class specific and should not be transferred to steel or aluminium components without independent validation. A direct quantitative comparison of the fitted exponents across material classes was outside the scope of the literature reviewed for this work and is suggested as a useful direction for future studies.

Limitations

Several limitations of the present study should be noted. The fit rests on only ten data points for a two-parameter nonlinear regression, so the parameters a and b carry non-negligible statistical uncertainty that was not quantified here (e.g. via bootstrap confidence intervals); a larger specimen set would allow such an analysis. All experimental data were obtained under fully axial, constant-amplitude loading at a single mean-stress level; mean-stress and multiaxial effects, surface finish, and size effects beyond the tested specimen diameters were not investigated and may shift the support factor for components outside the tested range. Finally, the R_m values used for the Group 6 specimens were read from staircase chart legends rather than from independent tensile tests, which introduces an additional, unquantified source of input uncertainty that propagates directly into the fitted exponent on R_m.

The sensitivity of the model to this last source of uncertainty can be estimated directly from the model form: since $n - 1 = a \cdot \sqrt{G} \cdot R_m^{b}$, a relative error δ in R_m propagates to a relative error of approximately $b \cdot \delta$ in $(n - 1)$. For $b = -0.4659$, a $\pm 10\%$ uncertainty in R_m therefore changes the predicted support increment $(n - 1)$ by only about $\mp 4.7\%$; for G6-D ($n - 1 = 0.327$), this corresponds to a shift in n of roughly ± 0.015, i.e. from about 1.31 to 1.34. The model is thus comparatively insensitive to moderate uncertainty in the R_m input, which partially mitigates the limitation noted above.

In conclusion, a practical two-parameter analytical model, $n = 1 + 3.7010 \cdot \sqrt{G} \cdot R_m^{-0.4659}$, has been derived and validated for the support factor of notched copper alloy components under axial loading, achieving an average prediction error of about 6.3% against the combined experimental and literature dataset. The model, together with the accompanying design charts, allows the fatigue-relevant support factor to be estimated directly from component geometry and material tensile strength, reducing the need for additional component-level fatigue testing.

Outlook

Future work should extend the experimental dataset with additional copper alloy specimens, particularly in the intermediate stress-gradient range ($1 < G < 2$ mm^{-1}) where the current dataset is sparse, and should include independently measured tensile strengths rather than chart-legend estimates. Validating the model under bending and torsional loading, and against a wider set of copper alloy grades (e.g. high-strength bronzes and brasses), would clarify whether the fitted exponents are specific to axial loading of the tested alloys or generalise more broadly. Finally, a formal uncertainty analysis of the fitted parameters a and b, e.g. via bootstrap resampling of the ten data points, would allow confidence bounds to be attached to the design charts in Figures 2 and 3, which would be of direct practical value for conservative component design.

References

[1] Siebel, E.; Stieler, M.: Ungleichförmige Spannungsverteilung bei schwingender Beanspruchung. VDI-Z. 97 (1955), pp. 121–152.

[2] DIN 50100:2016-12, Schwingfestigkeitsversuch – Durchführung und Auswertung von zyklischen Versuchen mit konstanter Lastamplitude für metallische Werkstoffproben und Bauteile.

[3] Rennert, R.; Kullig, E.; Vormwald, M.; Esderts, A.; Luke, M.: Rechnerischer Festigkeitsnachweis für Maschinenbauteile aus Stahl, Eisenguss und Aluminiumwerkstoffen. 7th ed., VDMA-Verlag, Frankfurt/Main, 2020.

[4] Virtanen, P.; Gommers, R.; Oliphant, T. E.; et al.: SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods 17 (2020), No. 3, pp. 261–272.

[5] Lukáš, P.; Kunz, L.; Svoboda, M.: Fatigue notch sensitivity of ultrafine-grained copper. Materials Science and Engineering A 391 (2005), No. 1–2, pp. 337–341.